Output of Router 0

Router#show ip router Static

S 30.0.0.0/8 [10/0] via 20.0.0.2

S 30.0.0.100/32 [10/0] via 40.0.0.2

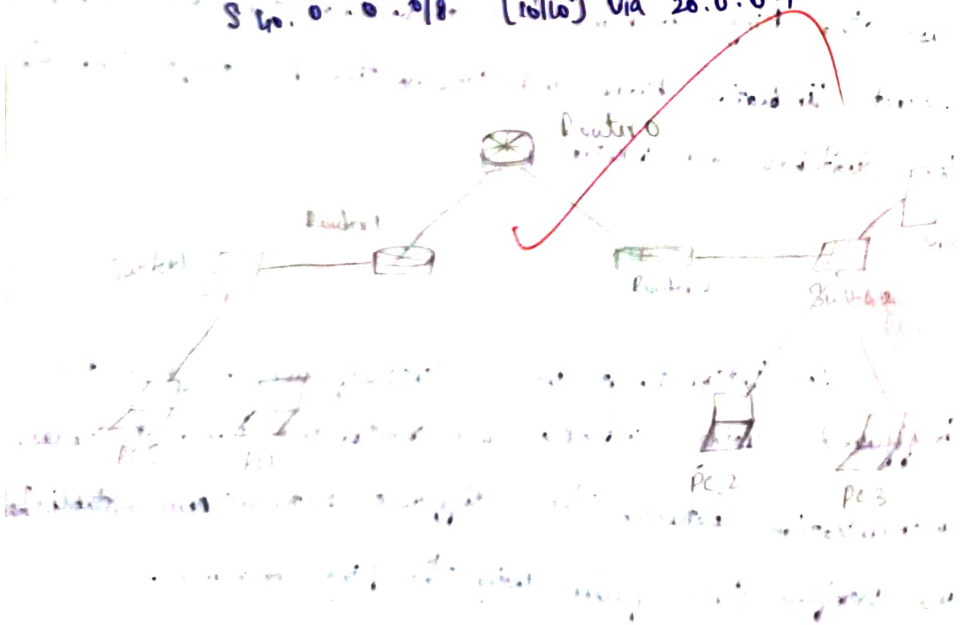
S 60.0.0.0/8 [10/0] via 40.0.0.2

Router 1

Router # show ip route Static

S 10.0.0.0/8 [10/0] via 20.0.0.1

S 40.0.0.0/8 [10/10] via 20.0.0.1



Routing at Network Layer

a) Simulates Static Routing Protocol Configuration using Cisco Packet Tracer

Aim: To Configure Static routing between different network using Cisco packet Tracer and verify communication between the network.

Steps:

1. Open Cisco packet Tracer.
2. Place 2 routers, 2 Switches and PC's.
3. Connect PC's to Switches and Switches to routers using Cables
4. Assign IP addresses to all PC's and router interfaces.
5. Configure the router interface and enable the using no shutdown
6. Configure Static routers using the Command
7. Configure Static routers on both routers.
8. Set the appropriate default gateway on each PC.
9. Use the Ping Command to test Communication between diffe network
10. Use Simulation Mode to observe the packet path.

Result:

That the Static routing was Successfully configure between the network using Cisco packet tracer and Communication between the networks was Successfully verified using the Ping Command.

Ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data.

Request timed out:

Reply from 10.0.0.2 : bytes = 32 time = 3ms TTL = 128

Reply from 10.0.0.0.2 : bytes = 32 time = 3ms TTL = 126

Reply from 10.0.0.2 bytes = 32 time = 3ms TTL = 126

Ping Statistics for 10.0.0.2

Packets: Sent = 4, Received = 3, Lost = 1

Approximate round trip in mili-seconds

Minimum = 3ms, Maximum = 3ms, Average = 3ms.

11.b) Simulate RIP using Cisco packet Tracer.

Aim: To configure and Simulate the Routing Information Protocol (RIP) between routers using Cisco packet Tracer and verify network Connectivity.

Steps:

1. Open Cisco packet Tracer.
2. place 2 or 3 routers, Switches and PCs.
3. Connect the devices using Ethernet Cables.
4. Assign IP addresses to PC's and router interfaces.
5. Enable all router interfaces using no shutdown.
6. Configure RIP on the Routers.
7. Configure the appropriate network addresses on each router according to the topology.
8. Check the routing table using: Show ip route.
9. Use the ping command to test Communication between PC's in different networks.
10. Use Simulation Mode to Observe how RIP learns and Select routers.

Result:

RIP was Successfully Configured between the routers using Cisco packet Trace. The routes learned the required network routers dynamically and Communication between different networks was Successfully verified.